Test Series Code

11

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | **DROPPER** | **M. MARKS: 720**

DATE: 05/04/2026

SYLLABUS

Physics:	Full Syllabus * As per NTA
Chemistry:	Full Syllabus * As per NTA
Botany:	Full Syllabus * As per NTA
Zoology:	Full Syllabus * As per NTA

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

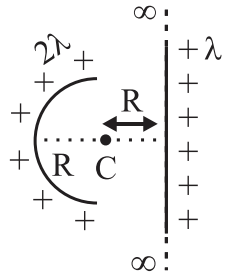
1. Three lenses in contact have a combined focal length of 20 cm. If the third lens is removed, the combined focal length becomes 15 cm. The third lens is:
(1) A converging lens of focal length 60 cm
(2) A diverging lens of focal length 60 cm
(3) A diverging lens of focal length 30 cm
(4) A converging lens of focal length 30 cm

2. A uniform rod of length 1.2 m stands vertically on a floor. It falls over without slipping. The speed of the top end when it hits the floor is ($g = 10$ m/s^2):
(1) 6.0 m/s (2) 4.8 m/s
(3) 8.2 m/s (4) 3.5 m/s

3. A constant torque acting on a flywheel increases its angular momentum from $2L_0$ to $6L_0$ in 2 seconds. The magnitude of the torque (in N-m) is:
(1) L_0 (2) $4L_0$
(3) $3L_0$ (4) $2L_0$

4. An increase in the temperature of a gas filled in a closed container would lead to:
(1) An increase in its mass
(2) An increase in its kinetic energy
(3) A decrease in its pressure
(4) A decrease in intermolecular distance

5. In the figure shown, a semicircular arc of radius R has a uniform linear charge density $+2\lambda$. An infinitely long straight wire with a uniform linear charge density $+\lambda$ is placed at a distance R from the center C of the arc in the same plane as the arc. The net electric field at point C is:



(1) $\frac{\lambda}{2\pi\epsilon_0 R}$ towards the right
(2) $\frac{\lambda}{2\pi\epsilon_0 R}$ towards the left
(3) $\frac{3\lambda}{2\pi\epsilon_0 R}$ towards the left
(4) Zero

6. A metal wire stretches by 4 mm when a load of 200 N is suspended from it. The elastic potential energy stored in the stretched wire is:
(1) 0.8 J (2) 0.4 J
(3) 400 J (4) 800 J

7. An electron is accelerated from rest through a potential difference of 100 V. The de Broglie wavelength associated with the electron is approximately:
(1) 1.23 nm (2) 12.3 nm
(3) 0.123 nm (4) 123 nm

8. A deuteron and an alpha particle enter a uniform magnetic field B perpendicularly with the same momentum. The ratio of the radii of their circular paths $r_d : r_\alpha$ is:
(1) 1: 2 (2) 2: 1
(3) 1: 1 (4) 4: 1

9. A variable force $F = (3y + 4)$ N acts on a particle in the y-direction. The work done by this force to displace the particle from $y = 0$ to $y = 2$ m is:
(1) 14 J (2) 10 J
(3) 20 J (4) 12 J

10. A block of mass 5 kg is held against the inner wall of a hollow cylinder of radius 2 m. The coefficient of friction between the block and the wall is 0.2. What is the minimum angular velocity (ω) required to keep the block from sliding down? ($g = 10$ m/s^2)
(1) 2 rad/s (2) 5 rad/s
(3) 10 rad/s (4) 25 rad/s

11. In an interference experiment with two coherent sources, the visibility of the fringes (also known as fringe contrast) is defined by the ratio $\frac{I_{max} - I_{min}}{I_{max} + I_{min}}$. If this value is found to be 0.8, what is the ratio of the maximum intensity to the minimum intensity $\left(\frac{I_{max}}{I_{min}}\right)$ in the resulting pattern?
(1) 4 (2) 9
(3) 16 (4) 25

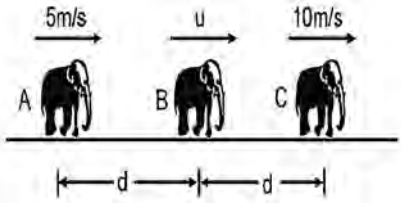
12. A defective balance with unequal arms weighs a body as 9 kg in one pan and 25 kg in the other. The true weight of the body is:
(1) 15 kg (2) 17 kg
(3) 34 kg (4) 16 kg

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13. Mass A collides elastically head on with mass B which is at rest. After the collision, mass A rebounds directly back with $\frac{1}{3}$ of its initial speed. The ratio of masses $m_B : m_A$ is:

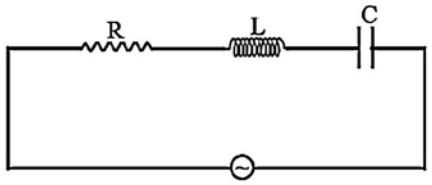
(1) 4: 1 (2) 3: 1
(3) 1: 2 (4) 2: 1

14. Three elephants A, B, and C are moving along a straight line with constant speeds in the same direction. Speed of A is 5 m/s and speed of C is 10 m/s. Initially, the separation between A & B is 'd' and between B & C is also 'd'. When 'B' catches 'C', the separation between A & C becomes 3d. Find the speed of B (in m/s).



(1) 15 (2) 20
(3) 5 (4) 25

15. In the LCR circuit shown, the RMS voltages across the resistor, inductor, and capacitor are equal. If the applied voltage is $v = 200\sin(100t)$ volts (where t is in seconds), the RMS value of voltage across the resistor will be:



$V = 200 \sin(100\ t)$

(1) 200 V (2) $100\sqrt{2}$ V
(3) 100 V (4) $200\sqrt{2}$ V

16. Regarding a meter bridge experiment, consider the following statements:

Statement I: For a proper circuit connection, the galvanometer deflection must be in opposite directions when the jockey touches the two endpoints of the wire.

Statement II: The jockey should be dragged along the wire to quickly find the null point.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

17. The velocity v of a particle at time t is given by the equation: $v = at + \frac{b}{t+c}$. Match the constants in **List-I** with their respective dimensional formulas in **List-II**:

List-I (Constant)		List-II (Dimensions)	
(A)	a	(P)	[T]
(B)	b	(Q)	$[LT^{-2}]$
(C)	c	(R)	[L]

(1) A-P, B-Q, C-R (2) A-Q, B-R, C-P
(3) A-R, B-P, C-Q (4) A-Q, B-P, C-R

18. To a cyclist moving North at a speed of 5 km/h, a truck appears to move with a speed of 13 km/h. If the truck is actually traveling due East, what is its actual speed?

(1) 18 km/h (2) 8 km/h
(3) 12 km/h (4) 10 km/h

19. A particle initially located at coordinates (2, 5) m starts from rest with a constant acceleration of $\vec{a} = 6\hat{j}$ m/s^2. Its position coordinates after 2 seconds will be:

(1) (2,17) (2) (14,5)
(3) (2,11) (4) (8,17)

20. A particle executes simple harmonic motion (SHM) with amplitude A. Let t_1 be the time taken to travel from the mean position ($x = 0$) to $x = A/2$, and t_2 be the time taken to travel from $x = A/2$ to the extreme position ($x = A$). The correct relation between t_1 and t_2 is:

(1) $t_1 = t_2$ (2) $t_2 = 2t_1$
(3) $t_1 = 2t_2$ (4) $t_2 = 3t_1$

21. A travelling wave is described by the equation $y = 5\sin(100\pi t - 0.5x)$, where x and y are in meters and t is in seconds. The frequency of the wave is:

(1) 100 Hz (2) 25 Hz
(3) 200 Hz (4) 50 Hz

22. A tuning fork A of frequency 300 Hz produces 5 beats per second with tuning fork B. When fork B is loaded with a small piece of wax, the beat frequency decreases to 2 beats per second. The original frequency of fork B was:

(1) 295 Hz (2) 310 Hz
(3) 305 Hz (4) 300 Hz

23. One mole of a diatomic gas is contained in a vessel. If 50% of the gas dissociates into individual atoms, the new average degree of freedom of the mixture will be nearest to: (neglect vibrational modes)

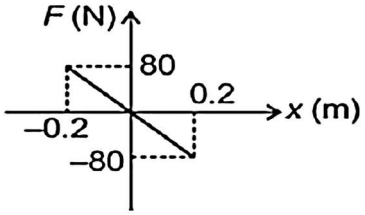
(1) 3.67 (2) 4.5
(3) 5.23 (4) 2.8

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24. The energy gap for a specific transition in a semiconductor is 3.1 eV. Using $hc = 1240$ eV-nm, the maximum wavelength of light capable of inducing this transition is:

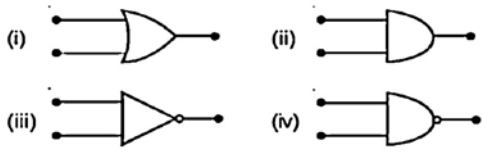
(1) 620 nm (2) 400 nm
(3) 310 nm (4) 200 nm

25. A body of mass 0.01 kg executes simple harmonic motion (SHM) about $x = 0$ under the influence of a restoring force as shown in the force-displacement (F vs x) graph below. The time period of the SHM is nearly:



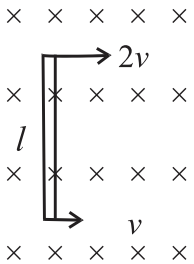
(1) 1.05 s (2) 0.52 s
(3) 0.25 s (4) 0.03 s

26. Symbolic representations of four logic gates are shown below. Pick out which ones represent AND, NAND, and NOT gates, respectively:



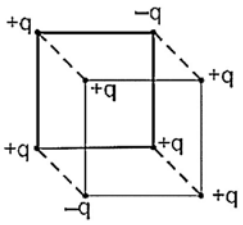
(1) (ii), (iv) and (iii) (2) (ii), (iii) and (iv)
(3) (iii), (ii) and (i) (4) (iii), (ii) and (iv)

27. A rod of length l is moving in a uniform magnetic field B (pointing into the page). The bottom-most point has a velocity v and the top-most point has a velocity $2v$. The emf induced across the endpoints of the rod is:



(1) Bvl (2) $2Bvl$
(3) $\frac{3}{2}Bvl$ (4) $\frac{1}{2}Bvl$

28. A cube has a side of $\sqrt{3}$ cm. Point charges are kept at each corner as shown in the diagram. Find the interaction electrostatic potential energy of the system (in Joules) when q = 10 μC.

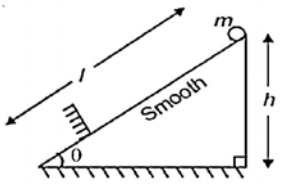


(1) 30 J (2) 100 J
(3) 120 J (4) Zero

29. A small repair tool is "dropped" (released gently without any additional push) from a satellite that is in a stable circular orbit around the Earth. Neglecting air resistance at that altitude, what will happen to the tool?

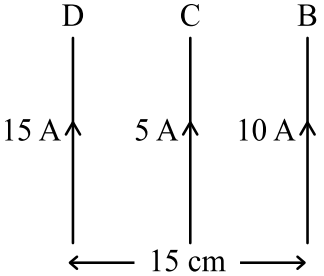
(1) It will fall vertically downward in a straight line toward the Earth's surface.
(2) It will fly off in a straight line away from the Earth into deep space.
(3) It will continue to move in the same orbit as the satellite.
(4) It will gradually slow down and spiral into the Earth's atmosphere within a few minutes.

30. A ball of mass m is released from the top of a smooth inclined plane of height h and length l. It strikes a rigid surface elastically at a distance of $\frac{3l}{4}$ from the top as shown. The impulse imparted to the ball by the rigid surface is:



(1) $m\sqrt{\frac{3}{2}gh}$ (2) $m\sqrt{3gh}$
(3) $2m\sqrt{3gh}$ (4) $m\sqrt{6gh}$

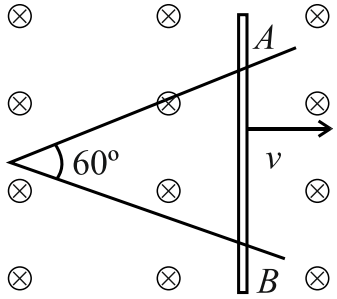
31. Three long, straight, and parallel wires carrying currents are arranged as shown. Wire C carries a current of 5 A and is placed such that it experiences no net force. The distance of wire C from wire D is:



(1) 9 cm (2) 7 cm
(3) 5 cm (4) 3 cm

[Contd...

32. A rod AB slides on a V-shaped wire with speed v as shown, such that at any time $OA = OB = l$. The magnetic field in the region is perpendicular downwards with strength B. The induced emf in the rod is:



(1) Zero (2) Bvl

(3) $\frac{\sqrt{3}Bvl}{2}$ (4) $\frac{Bvl}{2}$

33. Electromagnetic (EM) radiations are caused by:
(1) Stationary charges
(2) Uniformly moving charges
(3) Accelerated charges
(4) All of these

34. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Nuclear density is almost the same for all nuclei.
Reason R: The radius (r) of a nucleus depends on the mass number (A) as $r \propto A^{1/3}$.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is NOT the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

35. A point charge of $5\mu C$ is placed at the center of a solid sphere of radius 15 cm. The same charge is then placed at the center of a hollow cube of side 30 cm. The ratio of the electric flux emerging from the sphere to that of the cube is:
(1) 1: 8 (2) 1: 1
(3) 4: 3 (4) 1: 2

36. A lightning bolt strikes the ground with a voltage of 5×10^6 V. If the strike delivers a total charge of 10 C over a duration of 0.1 s, the average power of the lightning strike is:
(1) 50 MW (2) 500 MW
(3) 250 MW (4) 5 GW

37. A metallic wire of resistance 20 Ω is stretched to reduce its radius to half of its original value. The new resistance of the wire will be:
(1) 80 Ω (2) 160 Ω
(3) 640 Ω (4) 320 Ω

38. A planet has a density 4 times that of the Earth, but the acceleration due to gravity (g) on its surface is twice that of Earth. If the radius of the Earth is R, the radius of the planet is:
(1) $\frac{R}{2}$ (2) 2R
(3) $\frac{R}{4}$ (4) 4R

39. The work done to expand the radius of a soap bubble from 1 cm to 3 cm is $6.4\pi\times10^{-4}$ J. The surface tension of the soap solution is:
(1) 0.05 N/m (2) 0.2 N/m
(3) 0.1 N/m (4) 0.4 N/m

40. On a faulty thermometer scale, the freezing point of water (0° C) reads as 20 cm and the boiling point (100° C) reads as 120 cm. If the scale reading is 70 cm, the actual temperature is:
(1) 40° C (2) 50° C
(3) 60° C (4) 45° C

41. A spherical black body radiates 200 W of power at temperature T. If the radius of the sphere is doubled and the temperature is halved, the new radiated power will be:
(1) 50 W (2) 100 W
(3) 25 W (4) 400 W

42. A man has mass 4 times that of a boy. Initially, they have the same kinetic energy. If the man increases his speed by 5 m/s, his kinetic energy becomes 4 times that of the boy. The original speed of the man is:
(1) 2.5 m/s (2) 10 m/s
(3) 5 m/s (4) 7.5 m/s

43. A magnetic needle with magnetic moment 0.5 A m^2 is placed in a uniform magnetic field of 0.2 T. The work done to rotate the magnet from its stable equilibrium to its unstable equilibrium is:
(1) 0.1 J (2) 0.2 J
(3) -0.1 J (4) 0.05 J

44. For a hydrogen-like atom with atomic number $Z = 9$, which quantum state (n) has the same orbital radius as the ground state of the Hydrogen atom?
(1) $n = 1$ (2) $n = 9$
(3) $n = 3$ (4) $n = 2$

45. A vertical stick of length 36 cm stands on the surface of a lake ($\mu = 4/3$). To a fish underwater, vertically below the stick, the tip of the stick appears to be:
(1) 27 cm above the surface
(2) 36 cm above the surface
(3) 48 cm above the surface
(4) 52 cm above the surface

[Contd...

46. For the reaction, rate is expressed as:
Rate = $k_1[(CH_3)_2CHCl] + k_2[(CH_3)_2CHCl][OH^-]$
$(CH_3)_2CH–Cl + NaOH \rightarrow (CH_3)_2CH–OH$
(1) The reaction proceeds by S_N1 route only.
(2) The reaction proceeds by S_N2 route only.
(3) The reaction proceeds by S_N1 as well as by S_N2 routes (mixed mechanism).
(4) If concentration of RX is doubled, then rate of reaction will increase four times.

47. Elimination reaction of 2-chlorobutane to form but-2-ene is
A. β-Elimination reaction
B. Follows Zaitsev rule
C. Dehydrohalogenation reaction
D. Dehydration reaction
(1) A, C, D (2) B, C, D
(3) A, B, D (4) A, B, C

48. The molecular formula of second homologue in the homologous series of monocarboxylic acids is:
(1) $C_2H_4O_2$ (2) $C_3H_6O_2$
(3) CH_2O_2 (4) $C_2H_2O_2$

49. Which of the following is **correctly** matched?
(1) Basic oxides ⇒ Na_2O, CaO, SO_3
(2) Neutral oxides ⇒ CO_2, NO_2, N_2O
(3) Acidic oxides ⇒ ZnO, SO_2, P_2O_5
(4) Amphoteric oxides ⇒ Al_2O_3, ZnO, SnO_2

50. The number of moles of hydrogen molecules required to produce 12 moles of ammonia through Haber's process is:
(1) 6
(2) 18
(3) 24
(4) 36

51. $Mg^{2+} + NH_3 + HPO_4^{2-} \rightarrow X$ compound X is:
(1) $MgNH_4PO_4$
(2) $Mg_3(PO_4)_2$
(3) Both (1) and (2)
(4) $Mg(OH)_26H_2O$

52. Which of the following will **not** give tests for all the ions present in its aqueous solution?
(1) $FeCl_3$
(2) $K_4[Fe(CN)_6]$
(3) $(NH_4)_2Fe(SO_4)_2 \cdot 6H_2O$
(4) $KAl(SO_4)_2 \cdot 12H_2O$

53. Which amongst the following reactions of alkyl halides produces isonitrile as a major product?
A. $R–X + HCN \rightarrow$ B. $R–X + AgCN \rightarrow$
C. $R–X + KCN \rightarrow$ D. $R–X + NaCN \rightarrow$
Choose the *most appropriate* answer from the options given below:
(1) D only (2) C and D only
(3) B only (4) A and B only

54. The pair in which both species have the same magnetic moment (spin-only value) is:
(1) $[Fe(H_2O)_6]^{3+}$ and $[Mn(H_2O)_6]^{2+}$
(2) $[Co(H_2O)_6]^{2+}$ and $[Ni(H_2O)_6]^{2+}$
(3) $[Cr(H_2O)_6]^{3+}$ and $[Fe(H_2O)_6]^{2+}$
(4) $[Mn(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$

55. Calculate wavelength for the 3rd line of Balmer series of He^+ ion.
(1) $\frac{20}{21R}$ (2) $\frac{21}{25R}$
(3) $\frac{25}{21R}$ (4) $\frac{4}{3R}$

56. One of the commonly used reference electrodes is calomel electrode. It belongs to:
(1) Metal–insoluble salt–anion electrode
(2) Oxidation–reduction electrode
(3) Gas–ion electrode
(4) Metal ion–metal electrode

57. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: In aqueous solution, Fe^{2+} is reducing while Fe^{3+} is oxidising in nature.
Reason R: Half-filled and fully filled subshells are relatively more stable than partially filled subshells.
In the light of the above statements, choose the *most appropriate* answer:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is false but R is true.
(4) A is true but R is false.

58. Three aqueous solutions of KCl are labelled A, B and C with concentrations 0.2 M, 0.02 M and 0.002 M respectively. The van't Hoff factor follows:
(1) $i_A < i_B < i_C$ (2) $i_A > i_B > i_C$
(3) $i_A = i_B = i_C$ (4) $i_A < i_B > i_C$

[Contd...

59. Among the following species, select the diamagnetic molecule:

(1) B_2 (2) CO

(3) NO (4) O_2

60. Which of the following oxides is basic in nature?

(1) Na_2O (2) CO_2

(3) Al_2O_3 (4) P_2O_5

61. What is the product in the following reaction?

$$H_2C{=}CH{-}\underset{\displaystyle CH_3}{\underset{|}{CH}}{-}OH \xrightarrow[\Delta]{MnO_2}$$

(1) $H_2C{=}CH{-}\underset{\displaystyle CH_3}{\underset{|}{C}}{=}O$ (2) $OHC{-}\underset{\displaystyle CH_3}{\underset{|}{C}}{=}O$

(3) $HOOC{-}\underset{\displaystyle CH_3}{\underset{|}{C}}{=}O$ (4) CH_3CHO

62. The uncertainty in momentum of an electron is 2.0×10^{-24} kg m s^{-1}. The uncertainty in position will be: ($h = 6.626 \times 10^{-34}$ J s)

(1) 2.64×10^{-11} m (2) 5.27×10^{-11} m

(3) 1.32×10^{-10} m (4) 5.27×10^{-10} m

63. For which reaction from the following, ΔS will be maximum?

(1) $Ca(s) + \frac{1}{2}O_2(g) \rightarrow CaO(s)$

(2) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

(3) $C(s) + O_2(g) \rightarrow CO_2(g)$

(4) $N_2(g) + O_2(g) \rightarrow 2NO(g)$

64. Select the structure with **correct** numbering in the chain:

(1) $\overset{5}{C}H_2{=}\overset{4}{C}H{-}\overset{3}{C}H_2{-}\overset{2}{C}{\equiv}\overset{1}{C}H$

(2) $\overset{1}{C}H_3{-}\overset{2}{C}H{=}\overset{3}{C}H{-}\overset{4}{C}{\equiv}\overset{5}{C}H$

(3) $\overset{7}{C}H_2{=}\overset{6}{C}H{-}\overset{5}{C}H{=}\overset{4}{C}H{-}\overset{3}{C}H_2{-}\overset{2}{C}H{=}\overset{1}{C}H_2$

(4) $\overset{1}{C}H_2{=}\overset{2}{C}H{-}\overset{3}{C}H{=}\overset{4}{C}H{-}\overset{5}{C}H_2{-}\overset{6}{C}{\equiv}\overset{7}{C}H$

65. For the reaction A → Products, if $k = 2 \times 10^{-3}$ s^{-1}, the value of $-\frac{d[A]}{dt}$ when [A] = 3 M is:

(1) 3×10^{-3} mol L^{-1} s^{-1}

(2) 6×10^{-3} mol L^{-1} s^{-1}

(3) 9×10^{-3} mol L^{-1} s^{-1}

(4) 12×10^{-3} mol L^{-1} s^{-1}

66. Which of the following is **not** a state function of a thermodynamic system?

(1) Enthalpy (H) (2) Internal energy (U)

(3) Heat (q) (4) Gibbs free energy (G)

67. Aluminium does **not** react with pure water because:

(1) It is a soft metal

(2) It is covered by a thin oxide layer

(3) It has low density

(4) It is less electropositive than magnesium

68. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Λ_m	(I)	S cm^{-1}
(B)	E_{cell}	(II)	m^{-1}
(C)	κ	(III)	S cm^2mol^{-1}
(D)	G*	(IV)	V

Choose the **correct** answer from the options given below:

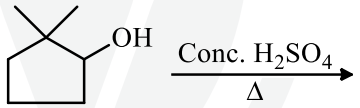
(1) A-II, B-III, C-IV, D-I

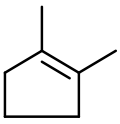
(2) A-II, B-IV, C-III, D-I

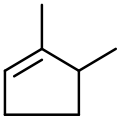
(3) A-III, B-IV, C-I, D-II

(4) A-II, B-I, C-III, D-IV

69. Identify the major product.



(1)  (2) 

(3)  (4) 

70. Transition elements act as catalysts because:

(1) they have high melting points

(2) they have high ionisation enthalpy

(3) they have high density

(4) they can show variable oxidation states

71. Vapour pressure of a solution containing 9 g non-volatile solute in 90 g water is 36 Torr. On adding 1 mole water, vapour pressure increases by 3 Torr. Vapour pressure of pure water is:

(1) 39 Torr

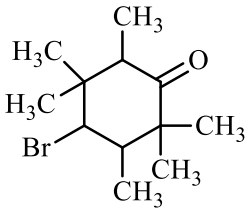
(2) 66.9 Torr

(3) 42 Torr

(4) 45 Torr

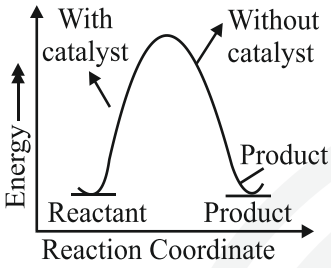
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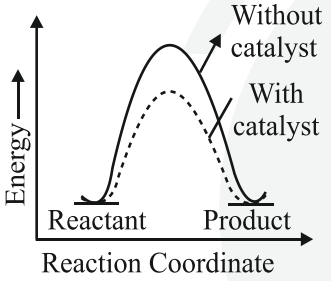
72. Examine the compound below. How many stereoisomers having this constitution are possible?

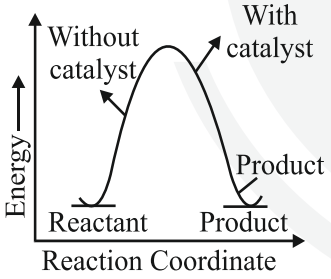


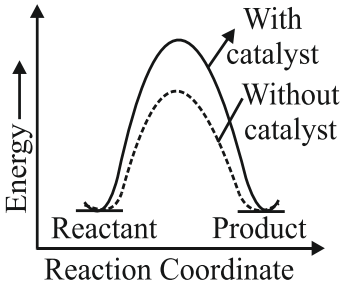
(1) 2 (2) 4
(3) 6 (4) 8

73. The **correct** reaction profile diagram for a negative catalyst reaction.

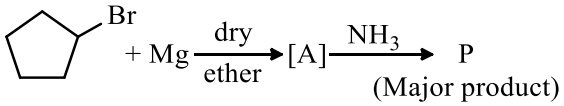
(1)


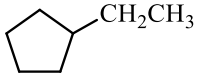
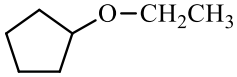
(2)


(3)


(4)


74. In the following sequence of reactions, the P is:



(1)  (2) 

(3)  (4) 

75. Which of the following have the same number of significant figures?

(A) 0.0064 (B) 4.002
(C) 18.0 (D) 243

Choose the **correct** answer from the options given below.

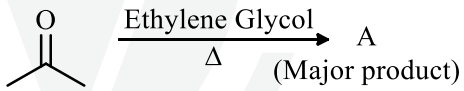
(1) A and B only (2) C and D only
(3) A, C and D only (4) B and C only

76. Match **List-I** with **List-II** for preparation of benzaldehyde.

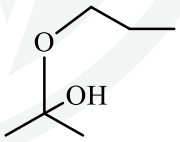
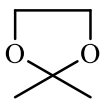
List-I		List-II	
(A)	Benzene	(I)	HCl and $SnCl_2$, H_3O^+
(B)	Benzonitrile	(II)	H_2, Pd-$BaSO_4$, S and quinoline
(C)	Benzoyl chloride	(III)	CO, HCl and $AlCl_3$
(D)	Toluene	(IV)	CrO_2Cl_2/H^+

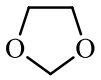
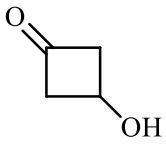
Choose the **correct** answer from the options given below:

(1) A-III, B-I, C-II, D-IV
(2) A-I, B-II, C-III, D-IV
(3) A-IV, B-III, C-I, D-II
(4) A-III, B-II, C-IV, D-I

77.


The product "A" in the above reaction is:

(1)  (2) 

(3)  (4) 

78. Using the same substituents, identify the **incorrect** statement(s):

(A) –OH is ortho-para directing.
(B) –CN is meta directing.
(C) Halides are deactivating but ortho-para directing.
(D) $-SO_3H$ is ortho-para directing.

Choose the **correct** answer:

(1) A and B only
(2) D only
(3) C and D only
(4) A only

[Contd...

79. For p-toluenesulphonyl chloride, which product is soluble in alkali during Hinsberg test?
(1) Product with tertiary amine
(2) Product with secondary amine
(3) Product with primary amine
(4) None of these

80. For SO_2, NO_2^- and N_3^-, identify the species in which the central atom is sp-hybridized:
(1) SO_2 only (2) NO_2^- only
(3) N_3^- only (4) SO_2 and NO_2^-

81. On boiling egg white, which protein structure is destroyed first among the following, while peptide sequence remains unchanged?
(1) Primary (2) Tertiary
(3) Peptide bond (4) Amide linkage only

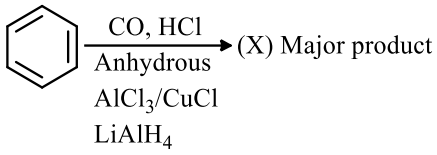
82. The mass of ammonia in grams produced when 2.8 kg of dinitrogen quantitatively reacts with 0.8 kg of dihydrogen is:
(1) 2800 g (2) 3000 g
(3) 3400 g (4) 5600 g

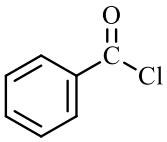
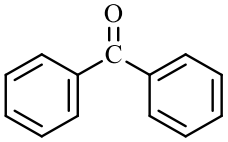
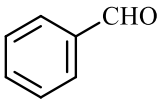
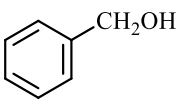
83. For elements B, C, N, Li, Be, O and F, the **correct** order of first ionization enthalpy is:
(1) Li < Be < B < C < N < O < F
(2) B < Li < Be > C > N > O > F
(3) Li < B < Be < C < O < N < F
(4) Li < Be < B < C < O < N < F

84. The semicarbazone formed from acetone contains how many nitrogen atoms?
(1) 1 (2) 2
(3) 3 (4) 4

85. Given below are two statements:
Statement I: Along a period, the chemical reactivity of elements increases uniformly from group 1 to group 18.
Statement II: Oxides of group 1 elements are generally basic, whereas oxides of many group 17 elements are acidic.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both Statement-I and Statement-II are true.
(2) Statement-I is true but Statement-II is false.
(3) Statement-I is false but Statement-II is true.
(4) Both Statement-I and Statement-II are false.

86. Identify major product "X" formed in the following reaction:



(1) 
(2) 
(3) 
(4) 

87. Among the following, which compound contains a –COOH group?
A. Sulphanilic acid B. Picric acid
C. Aspirin D. Ascorbic acid
(1) A only (2) C only
(3) A and C only (4) B, C and D only

88. The carbohydrate unit associated with DNA is asked about through these statements:
A. It is a pentose sugar
B. It is present in pyranose form
C. It is in "D" configuration
D. When free, it behaves as a reducing sugar
E. It is in α-anomeric form
Choose the **correct** answer:
(1) A, C and D only (2) B, D and E only
(3) A, B and E only (4) A, D and E only

89. Identify the **incorrect** statements regarding group 15 elements:
A. Dinitrogen is a diatomic gas which acts like an inert gas at room temperature.
B. The common oxidation states of these elements are –3, +3 and +5.
C. Nitrogen has unique ability to form pπ–pπ multiple bonds.
D. The stability of +5 oxidation states increases down the group.
E. Nitrogen shows a maximum covalency of 6.
Choose the **correct** option:
(1) D and E only (2) A, C, E only
(3) B, D, E only (4) A, B, D only

90. Among SbF_5, PCl_5, BrF_5 and $XeOF_4$, which species show octahedral electron-pair arrangement around the central atom?
(1) SbF_5 and PCl_5 (2) BrF_5 and $XeOF_4$
(3) PCl_5 and $XeOF_4$ (4) SbF_5 and BrF_5

[Contd...

91. With respect to treatment of sewage water, arrange the following steps in correct sequence.

A. Effluent from the primary settling tank is taken for secondary treatment.

B. Production of biogas.

C. Using activated sludge as inoculum in aeration tank.

D. Removal of soil and pebbles.

E. Release of treated water into natural water bodies.

Choose the **correct** option.

(1) B → E → D → C → A

(2) D → A → C → B → E

(3) B → D → C → A → E

(4) E → B → D → A → C

92. For the synthesis of fatty acids, which of the following will be withdrawn from the respiratory pathway?

(1) Amino acids

(2) PGAL

(3) Acetyl CoA

(4) Glucose

93. Match **List-I** with **List-II**.

List-I (Kingdom)		**List-II (General characteristic of members)**	
(A)	Monera	(I)	Unicellular eukaryotic
(B)	Protista	(II)	Multicellular autotrophic
(C)	Fungi	(III)	Unicellular prokaryotic
(D)	Plantae	(IV)	Multicellular heterotrophic

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-I, C-IV, D-II

(2) A-I, B-II, C-III, D-IV

(3) A-III, B-IV, C-I, D-II

(4) A-II, B-I, C-IV, D-III

94. An example of pleiotropy is;

(1) Haemophilia.

(2) Phenylketonuria.

(3) Colour blindness.

(4) Down's Syndrome.

95. Given below are two statements:

Statement I: The early stages of embryo development (embryogeny) are vastly different in monocotyledons and dicotyledons.

Statement II: In dicot plants, the zygote gives rise to the proembryo and subsequently to the globular, heart-shaped and mature embryo.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

96. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): A population growing in a habitat with limited resources shows an S-shaped growth curve.

Reason (R): A population growing in a habitat with limited resources show initially a log phase, followed by phases of acceleration and deceleration and finally an asymptote.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

97. Who amongst the following studied the effect of hard X-rays on bacteriophage multiplication?

(1) Francis Crick

(2) James Dewey Watson

(3) Har Gobind Khorana

(4) Marshall Nirenberg

98. In a maize grain, the aleurone layer;

(1) separates the embryo from the endosperm.

(2) stores food in the form of starch.

(3) encloses the plumule and radicle.

(4) is the outermost covering of the seed.

[Contd...

99. Identify the **incorrectly** matched pair.
(1) Mycoplasma - Lack a cell wall
(2) Euglenoids - Have a protein rich layer called pellicle
(3) Slime moulds - Form a plasmodium under suitable conditions
(4) Dinoflagellates - Cell wall has stiff chitinous plates

100. Females are diploid but males are haploid, in case of;
(1) fruitfly. (2) birds.
(3) honey bees. (4) grasshopper.

101. The growth rate is measured in terms of increase in cell number, for;
(1) dicot leaves.
(2) watermelon cell.
(3) pollen tube.
(4) maize root apical meristem.

102. During cell cycle, DNA replicates in one of the sub-stages of the;
(1) resting phase.
(2) most dramatic phase.
(3) prophase I of meiosis.
(4) karyokinesis of mitosis.

103. Which of the following sets of reactions occurs during the light reactions of photosynthesis?
(1) Light absorption, CO_2 fixation, oxygen release
(2) Water splitting, ATP formation, sugar synthesis
(3) Light absorption, water splitting, ATP and NADPH formation
(4) CO_2 fixation, sugar synthesis, regeneration of RuBP

104. Match **List-I** with **List-II** w.r.t the *lac* operon.

List-I (Gene)		**List-II (Product)**	
(A)	*y*	(I)	Repressor
(B)	*a*	(II)	*β*-galactosidase
(C)	*i*	(III)	Transacetylase
(D)	*z*	(IV)	Permease

Choose the *most appropriate* answer from the options given below.
(1) A-III, B-I, C-IV, D-II
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-I, C-IV, D-III

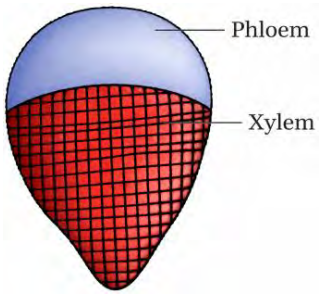
105. Which of the following **correctly** describes a taxonomic category?
(1) The basic unit of classification is the genus.
(2) Each rank or taxon represents a unit of classification.
(3) A particular organism can occupy only anyone taxonomic category.
(4) All taxonomic categories have the same number of organisms.

106. The type of population interaction between cuckoo and crow is;
(1) Mutualism. (2) Commensalism.
(3) Parasitism. (4) Predation.

107. Given below are two statements:
Statement I: Griffith, in his experiments, observed transformation of a virulent strain to non-virulent strain of pneumococcus.
Statement II: The Central dogma in molecular biology states that the genetic information flows from Protein → RNA → DNA.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

108. Select the correct statements regarding an ecosystem.
A. Identification and enumeration of plant species of an ecosystem gives its stratification.
B. Vertical distribution of different species occupying different levels is called species composition.
C. Productivity, decomposition, energy flow, and nutrient cycling are the four important functional components of an ecosystem.
D. In a pond ecosystem, decomposers like fungi, bacteria and flagellates are especially abundant at the top.
E. There is a unidirectional movement of energy towards the higher trophic levels and its dissipation as heat.
Choose the **correct** option:
(1) A and B only
(2) C and E only
(3) B, D and E only
(4) A, C and D only

109. The vascular bundle shown below is said to be;



(1) conjoint and closed.
(2) conjoint and open.
(3) radial and closed.
(4) radial and open.

110. During a meiotic cell cycle, a dyad of haploid cells is formed;
(1) by cytokinesis after telophase II.
(2) in zygotene.
(3) in pachytene.
(4) by cytokinesis after telophase I.

111. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The placentation in members of family Malvaceae is said to be axile.
Reason (R): A false septum is characteristic of ovary in members of family Malvaceae.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

112. The stroma of chloroplast contains all the following, **except**;
(1) ribosomes.
(2) chlorophyll pigments.
(3) enzymes for sugar synthesis.
(4) DNA.

113. Both chasmogamous and cleistogamous flowers are produced by all the following, **except**;
(1) papaya.
(2) *Viola.*
(3) *Commelina.*
(4) *Oxalis.*

114. The introduction of the African catfish *Clarias gariepinus* for aquaculture purposes is posing a threat to indigenous catfishes in our rivers. This is an example of:
(1) Co-extinction.
(2) Over-exploitation.
(3) Habitat loss.
(4) Alien species invasion.

115. Identify the **correct** physiological effect of gibberellins in plants.
(1) Apical dominance
(2) Phototropism
(3) Bolting
(4) Respiratory climactic

116. How many of the following show isogamous reproduction?
A. *Volvox*
B. *Spirogyra*
C. *Fucus*
D. *Ulothrix*
E. *Gelidium*
Choose the **correct** option.
(1) One only (2) Two only
(3) Three only (4) Four only

117. Select the **correct** statement about the endomembrane system.
(1) It includes the ER, golgi complex, mitochondria and vacuoles.
(2) It consists of organelles whose functions are coordinated.
(3) Peroxisomes and chloroplasts are part of the endomembrane system because they are membrane-bound.
(4) All organelles of the endomembrane system perform same function.

118. Choose the **correct** statements regarding post-transcriptional events in eukaryotes.
A. Tailing of hnRNA occurs at the 3'-end.
B. During splicing, introns are removed and exons are joined.
C. Functional mRNA is transported from the cytoplasm to the nucleus.
D. In capping, methyl guanosine bisphosphate is added to the 5'-end of hnRNA.
E. The primary transcript undergoes translation in nucleus.
Choose the **correct** option.
(1) A and B (2) B and C
(3) C and D (4) D and E

119. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Crossing over during meiosis is an enzyme-mediated process.

Reason (R): The enzyme recombinase leads to recombination of genetic material between two non-homologous chromosomes.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

120. _____ was/were awarded Nobel prize for mapping of the pathway of carbon assimilation in photosynthesis.

(1) Katherine Esau
(2) Melvin Calvin
(3) Hatch and Slack
(4) Jan Ingenhousz

121. Identify the **correctly** matched pair.

(1) Statin - Immunosuppressive agent
(2) Cyclosporin A - Blood-cholesterol lowering agent
(3) Streptokinase - Clot buster
(4) Lipase - Clarifying bottled juices

122. Match **List-I** with **List-II** w.r.t a typical angiosperm embryo sac at maturity.

List-I		List-II	
(A)	Synergids	(I)	Seven
(B)	Total cells	(II)	Two
(C)	Antipodals	(III)	One
(D)	Egg cell	(IV)	Three

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-I, C-IV, D-II
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-I, C-IV, D-III

123. Identify the correct statements regarding isobilateral leaves.

A. Stomata are present on both surfaces of the epidermis.
B. Mesophyll is differentiated into palisade and spongy parenchyma.
C. It is the characteristic of dicotyledonous plants.
D. Parallel venation is reflected in the near similar sizes of vascular bundles.
E. Bulliform cells are present in the adaxial epidermis of grasses.

Choose the **correct** option:

(1) A, D, E only
(2) B, C only
(3) A, C, D only
(4) B, D, E only

124. In a monohybrid cross between a true-breeding tall pea plant (TT) and a true-breeding dwarf pea plant (tt), what will be the genotypic ratio in the F_2 generation?

(1) 3 : 1
(2) 1 : 1
(3) 1 : 2 : 1
(4) 2 : 1 : 1

125. Given below are two statements:

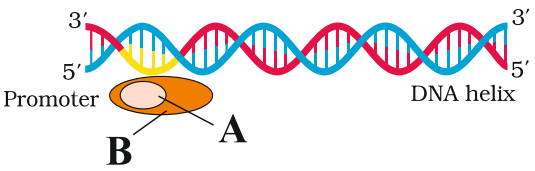
Statement I: *Marchantia* is a moss.

Statement II: *Polytrichum* is a liverwort.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

126. Shown below is the initiation step of transcription in bacteria. Labels A and B, respectively, are;



(1) rho factor and RNA polymerase.
(2) rho factor and DNA polymerase.
(3) sigma factor and RNA polymerase.
(4) sigma factor and DNA polymerase.

[Contd...

127. Given below are two statements:

Statement I: The prions are infectious agents consisting of abnormally folded DNA.

Statement II: The most notable disease caused by prions is bovine spongiform encephalopathy (BSE) in cattle.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

128. Identify the **true** statement.

(1) The products of aerobic respiration in yeast are CO_2 and ethanol.
(2) The products of aerobic respiration in bacteria are CO_2 and lactic acid.
(3) The products of anaerobic respiration in yeast are CO_2 and lactic acid.
(4) The product of anaerobic respiration in human muscle cells is lactic acid.

129. The conjuctive tissue in a dicot root is;

(1) the parenchymatous cells which lie between the xylem and the phloem.
(2) the sclerenchymatous tissue that provides mechanical support.
(3) a part of the endodermis containing casparian strips.
(4) the outermost layer of the stele.

130. The Z scheme of light reaction in photosynthesis involves:

(1) only PS I.
(2) only PS II.
(3) both PS I and PS II.
(4) neither PS I nor PS II.

131. The microvilli are;

(1) found in all eukaryotic cells and perform lysis of biomolecules.
(2) found in animal cells and help in absorption of digested food material.
(3) present in plants and connect the cytoplasm of neighbouring cells.
(4) are responsible for respiration in prokaryotic cells.

132. According to Robert May's global estimates, approximately what per cent of the total species have been recorded so far?

(1) 22%
(2) 50%
(3) >90%
(4) only 5%

133. Select the **incorrect** statement regarding the Grazing Food Chain.

(1) Producers belong to the first trophic level.
(2) Herbivores are primary consumers.
(3) It is the major conduit for energy flow in a terrestrial ecosystem.
(4) Carnivores that feed on herbivores are secondary consumers.

134. Match **List-I** with **List-II**.

List-I		**List-II**	
(A)	Pneumatophores	(I)	Support in sugarcane
(B)	Prop roots	(II)	Food storage in potato
(C)	Stilt roots	(III)	Support in Banyan tree
(D)	Underground stem	(IV)	Respiration in *Rhizophora*

Choose the *most appropriate* answer from the options given below.

(1) A-IV, B-III, C-I, D-II
(2) A-IV, B-I, C-III, D-II
(3) A-II, B-III, C-I, D-IV
(4) A-III, B-IV, C-II, D-I

135. Carrageen is obtained from certain marine;

(1) brown algae.
(2) green algae.
(3) blue-green algae.
(4) red algae.

[Contd...

136. Identify the **correct** statement regarding the different types of nitrogenous waste excretion:

(1) Ammonia is the most toxic form and requires a very small amount of water for its elimination.

(2) Uricotelic animals like reptiles and birds excrete nitrogenous waste as a paste or pellet with minimum loss of water.

(3) Ureotelic animals, such as mammals and terrestrial amphibians, excrete ammonia directly through their body surface.

(4) In aquatic insects and bony fishes, the kidneys play a significant role in the elimination of ammonia.

137. If a country shows declining MMR and IMR but no significant reduction in birth rate, the immediate demographic consequence would be:

(1) Stabilisation of population

(2) Negative population growth

(3) Rapid population expansion

(4) Decrease in proportion of reproductive-age individuals

138. Match **List-I** with **List-II**.

List-I		**List-II**	
(A)	Opioids	(I)	Has a potent stimulating action on central nervous system
(B)	Cannabinoids	(II)	Normally used as medicines to help patients cope with mental illnesses
(C)	Cocaine	(III)	Bind to specific opioid receptors present in our central nervous system and gastrointestinal tract
(D)	Barbiturates	(IV)	Obtained from the inflorescences of the plant *Cannabis sativa*

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I, D-IV

(2) A-III, B-II, C-I, D-IV

(3) A-III, B-IV, C-I, D-II

(4) A-III, B-II, C-IV, D-I

139. The formation of a large tumor mass in malignant cancer can indirectly cause death of nearby normal cells primarily due to:

(1) Secretion of digestive enzymes that lyse all surrounding cells

(2) Immediate immune destruction of adjacent tissues

(3) Activation of contact inhibition in normal cells

(4) Competition for vital nutrients by rapidly dividing neoplastic cells

140. Identify the **correct** statement regarding the chemical composition of living organisms:

(1) All elements present in the earth's crust are absent in living tissues.

(2) The relative abundance of carbon and hydrogen is higher in the earth's crust than in living organisms.

(3) The relative abundance of carbon and hydrogen is higher in any living organism than in the earth's crust.

(4) Living organisms are made of entirely different elements than those found in non-living matter.

141. Which region of the cerebral cortex is responsible for complex functions such as intersensory associations, memory, and communication?

(1) Motor area

(2) Sensory area

(3) Association areas

(4) Limbic system

142. Given below are two statements:

Statement I: Recombinant therapeutics do not induce unwanted immunological responses.

Statement II: PCR is used to detect mutations in genes in suspected cancer patients.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

143. In which of the following sets of animals do all members possess a coelom lined by mesoderm?

(1) Annelids, Molluscs, and Aschelminthes.

(2) Platyhelminthes, Echinoderms, and Chordates.

(3) Arthropods, Hemichordates, and Echinoderms.

(4) Coelenterates, Ctenophores, and Porifera.

[Contd...

144. Gout is a disorder of the skeletal system characterized by:
(1) Decreased bone mass and increased chances of fractures.
(2) Inflammation of joints due to accumulation of uric acid crystals.
(3) Wild contractions in muscle due to low Ca^{2+} in body fluid.
(4) Autoimmune inflammation of the synovial membrane.

145. Which of the following statements regarding placenta is/are **correct**?
A. Chorionic villi arise from the uterine tissue after implantation.
B. Placenta is formed by interdigitation of foetal chorionic villi and maternal uterine tissue.
C. Placenta functions only as a transport organ and has no endocrine role.
D. hCG and hPL are hormones specifically produced during pregnancy.
(1) B and D only (2) A, B and D only
(3) B, C and D only (4) A and C only

146. Which of the following statements is **correct** regarding polysaccharides?
(1) In a glycogen chain, the left end is the reducing end and the right end is the non-reducing end.
(2) Cellulose is a heteropolymer made of glucose and fructose.
(3) Cellulose forms helical secondary structures and can hold iodine molecules.
(4) Chitin is a complex polysaccharide found in the exoskeleton of insects and is a homopolymer.

147. Choose the **correctly** matched pair regarding the functions of different parts of the nephron:
(1) Proximal Convoluted Tubule (PCT) — Reabsorption of only 10% of electrolytes and water.
(2) Descending limb of Henle's loop — Impermeable to water but allows transport of electrolytes.
(3) Distal Convoluted Tubule (DCT) — Conditional reabsorption of Na^+ and water.
(4) Ascending limb of Henle's loop — Highly permeable to water, making the filtrate hypertonic.

148. Match **List-I** with **List-II.**

List-I		List-II	
(A)	*Pila*	(I)	Marine fish with ventral mouth
(B)	*Nereis*	(II)	Bioluminescent organism with external fertilisation
(C)	*Carcharodon*	(III)	File-like rasping organ called radula for feeding.
(D)	*Pleurobrachia*	(IV)	Dioecious aquatic form with lateral appendages called parapodia.

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-IV, C-II, D-I
(4) A-IV, B-I, C-II, D-III

149. Biopiracy refers to:
(1) Exploitation of bio-resources with proper authorisation or compensation
(2) Patenting GM crops without testing
(3) Exploitation of bio-resources without proper authorisation or compensation
(4) Use of viral vectors in gene therapy

150. Which of the following **correctly** justifies the need for medical consultation before adopting a contraceptive method?
(1) Contraceptives are universally free from side effects.
(2) Contraceptives may cause physiological side effects.
(3) Contraceptives permanently alter ovarian reserve.
(4) Contraceptives guarantee complete protection from all reproductive disorders.

151. Emphysema is a chronic disorder primarily characterized by:
(1) Proliferation of fibrous tissue in lungs leading to decreased respiratory surface area.
(2) Damage to alveolar walls, leading to decreased respiratory surface area.
(3) Inflammation of bronchi and bronchioles only.
(4) Spasms in the tracheal muscles causing wheezing.

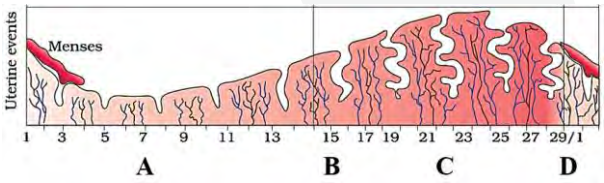
152. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Rheumatoid arthritis is classified as an auto-immune disease that leads to the damage of one's own body tissues.
Reason R: Auto-immunity arises exclusively due to genetic mutations that cause the immune system to lose its memory-based acquired immunity.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

153. Insertion of foreign DNA at *Bam*HI site in pBR322 will result in:
(1) Loss of ampicillin resistance only
(2) Loss of tetracycline resistance only
(3) Loss of both antibiotic resistances
(4) No change in resistance pattern

154. During the process of blood coagulation, which of the following conversions is catalyzed by the enzyme complex Thrombokinase?
(1) Conversion of soluble Fibrinogens into insoluble Fibrins.
(2) Conversion of Prothrombin into active Thrombin.
(3) Conversion of inactive thrombin into active prothrombin.
(4) Activation of platelets to release certain factors which initiate clotting.

155. Choose the **correctly** matched pair of secondary metabolite and its category:
(1) Concanavalin A — Alkaloid
(2) Ricin — Toxin
(3) Codeine — Essential oil
(4) Anthocyanin — Drug

156. Carefully observe the following diagram of uterine changes during menstrual cycle.



The hormonal condition most likely associated with stage D is:
(1) High estrogen and high progesterone
(2) LH surge
(3) Rapid decline in FSH
(4) Sudden decline in progesterone and estrogen

157. In the excretory and reproductive systems of a male frog:
(1) Ureters act as a urinogenital duct that opens directly into the urinary bladder.
(2) Vasa efferentia enter the kidneys and open into the Bidder's canal.
(3) The testes are adhered to the lower part of the kidneys by a fold of peritoneum called mesorchium.
(4) Copulatory pads are found on the first digits of the hind limbs.

158. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Parathyroid hormone (PTH)	(I)	Decreases blood glucose
(B)	Calcitonin	(II)	Increases blood calcium level
(C)	Insulin	(III)	Decreases blood calcium level
(D)	Aldosterone	(IV)	Promotes sodium reabsorption in kidney

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-I, D-IV
(2) A-II, B-III, C-IV, D-I
(3) A-II, B-III, C-I, D-IV
(4) A-III, B-IV, C-II, D-I

159. Regarding RNA interference (RNAi) to control nematodes in tobacco plants, identify the **incorrect** statement.
(1) RNAi occurs in all eukaryotic organisms as a cellular defence mechanism.
(2) RNAi prevents translation of specific mRNA.
(3) Both sense and antisense RNA are produced in host plant.
(4) dsRNA directly destroys nematode DNA.

160. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Exonuclease	(I)	Joins DNA fragments by forming phosphodiester bonds
(B)	Endonuclease	(II)	Removes nucleotides from the ends of DNA
(C)	DNA ligase	(III)	Specific sequence from where replication starts
(D)	Origin of replication	(IV)	Makes cuts at specific internal positions within DNA

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-II, B-IV, C-III, D-I
(3) A-II, B-IV, C-I, D-III
(4) A-III, B-I, C-IV, D-II

161. Choose the **correct** statement regarding *Periplaneta americana.*
(1) The presence of a boat-shaped 7th sternum forming a brood pouch in a male.
(2) A pair of short, thread-like anal styles arising from the 9th sternum in a female.
(3) Forewings called tegmina that are transparent and membranous, used primarily for flight.
(4) Mushroom-shaped glands located in the 6th to 7th abdominal segments in a male.

[Contd...

162. The diffusion membrane, through which the exchange of gases occurs, consists of how many major layers?
(1) Three: Squamous epithelium of alveoli, endothelium of capillaries, and the basement substance.
(2) Three: Ciliated epithelium of alveoli, endothelium of capillaries, and the basement substance.
(3) One: The fused membrane of alveoli and capillaries.
(4) Four: Including the pleural membrane and alveolar fluid.

163. Match **List-I** with **List-II.**

List-I (Part)		**List-II (No. of bones)**	
(A)	Skull (including hyoid and ear ossicles)	(I)	26
(B)	Vertebral column	(II)	24
(C)	Ribs	(III)	29
(D)	Hind limb (one leg)	(IV)	30

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-II, D-I
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-III, C-I, D-IV
(4) A-III, B-I, C-II, D-IV

164. Regarding evidence for evolution, read the following statements and choose **correct** ones.
I. Sweet potato and potato are examples of analogy.
II. Homology indicates common ancestry and divergent evolution.
III. Analogous structures are a result of convergent evolution.
IV. The eye of an octopus and a mammal are homologous organs.
(1) I and IV only (2) I, II, and III only
(3) II and III only (4) I, II, III, and IV

165. During recombinant DNA technology, maintenance of sterile conditions to grow only desired microbes is called:
(1) Genetic engineering
(2) Downstream processing
(3) Bioprocess engineering
(4) Gene cloning

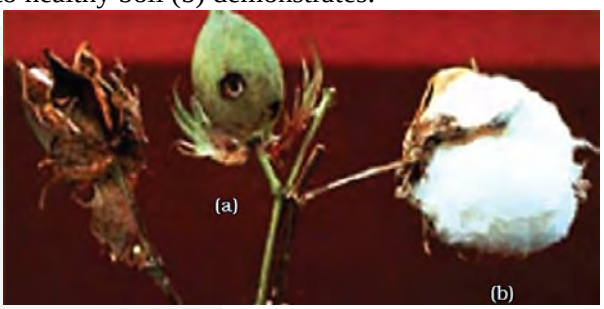
166. Match **List-I** with **List-II** regarding alimentary canal in cockroach.

List-I		**List-II**	
(A)	Crop	(I)	Ring of 6-8 blind tubules secreting digestive juice
(B)	Gizzard	(II)	Sac-like structure used for storing food
(C)	Hepatic caeca	(III)	100-150 yellow filamentous tubules for excretion
(D)	Malpighian tubule	(IV)	Has outer thick circular muscles with six chitinous teeth

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-III, D-IV
(2) A-II, B-IV, C-I, D-III
(3) A-II, B-IV, C-III, D-I
(4) A-IV, B-III, C-I, D-II

167. Which of the following conditions is essential for performing GIFT but not mandatory for IUT?
(1) Functional uterus
(2) At least one patent fallopian tube
(3) *In vitro* fertilisation facility
(4) Transfer of embryo beyond 8 blastomeres

168. In the figure, the damaged cotton boll (a) compared to healthy boll (b) demonstrates:



(1) Increased mineral uptake
(2) Effect of nematode infestation
(3) Viral infection in cotton
(4) Effect of bollworm infestation

169. A clinical analysis of a semen sample reveals a normal sperm count, but the sperm show significantly low motility. Which of the following is the most likely cause of this condition?
(1) Impaired spermatogenesis within the seminiferous tubules
(2) Failure in the process of spermiation from Sertoli cells
(3) Dysfunctional secretions from the epididymis and male accessory glands
(4) Inadequate secretion of Follicle Stimulating Hormone (FSH) from the anterior pituitary

170. Which of the following is **not** true regarding co-factors?
A. Apoenzyme is the non-protein portion of the enzyme.
B. Haem is a co-enzyme found in peroxidase and catalase.
C. NAD and NADP contain the vitamin Riboflavin.
D. Catalytic activity is lost when the co-factor is removed from the enzyme.
(1) A and C only
(2) D only
(3) A, B and C only
(4) A, B, C and D

171. Given below are two statements:
Statement I: Most of the cartilages in vertebrate embryos are replaced by bones in adults.
Statement II: Gap junctions are specialized for chemical communication but do not allow the transfer of small molecules.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

172. Identify the **correct** statement regarding the symmetry and development of specific animal groups:
(1) Adult echinoderms exhibit bilateral symmetry, while their larvae are radially symmetrical.
(2) Ctenophores are exclusively marine, diploblastic organisms that reproduce both asexually and sexually.
(3) In Aschelminthes, the female is often longer than the male, and the alimentary canal lacks a muscular pharynx.
(4) Sponges are mostly asymmetrical.

173. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The body of a reptile is covered by dry and cornified skin with epidermal scales or scutes.
Reason R: Most reptiles have a three-chambered heart, but crocodiles possess a four-chambered heart.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

174. Regarding the circulatory system of a frog, which of the following statements is **correct**?
(1) The heart is four-chambered and is covered by a membrane called pericardium.
(2) The renal portal system provides a venous connection between the liver and the intestine.
(3) Erythrocytes are nucleated and contain the red-colored pigment haemoglobin.
(4) The sinus venosus is a triangular structure that joins the left atrium.

175. During pregnancy, which of the following hormonal changes primarily contribute to maintenance of pregnancy and maternal metabolic adjustments?
(1) Decreased thyroxine and prolactin with increased cortisol
(2) Increased estrogens, progestogens, cortisol, prolactin and thyroxine
(3) Increased relaxin from pituitary during early implantation
(4) Decreased hCG after implantation leading to placental formation

176. The 'Lub' and 'Dub' heart sounds are produced due to the:
(1) Contraction of the atrial muscles.
(2) Closure of the heart valves.
(3) Flow of blood into the aorta.
(4) Opening of the semi-lunar valves**.**

177. Match **List-I** with **List-II**.

List-I		**List-II**	
(A)	Louis Pasteur	(I)	Proposed chemical evolution preceded life
(B)	Oparin & Haldane	(II)	Demonstrated life comes from pre-existing life
(C)	S.L. Miller	(III)	Created electric discharge in a closed flask at 800 ^{o}C
(D)	Hugo de Vries	(IV)	Proposed mutation as the cause of speciation

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-IV, D-III
(2) A-II, B-I, C-III, D-IV
(3) A-I, B-II, C-III, D-IV
(4) A-IV, B-II, C-I, D-III

178. Which of the following sets of diseases have been "controlled to a large extent" by the use of vaccines?
(1) Smallpox, Polio, and Dengue
(2) Diphtheria, Pneumonia, and Malaria
(3) Polio, Diphtheria, Pneumonia, and Tetanus
(4) Polio, Smallpox, and Chikungunya

179. Why is a thermostable DNA polymerase (*Taq* polymerase) required in PCR?
(1) It synthesises RNA primers.
(2) It denatures during annealing.
(3) It prevents primer annealing.
(4) It remains active during high-temperature denaturation.

180. During an acute stress situation, which of the following combinations of physiological changes is most directly mediated by catecholamines?
(1) Decreased heart rate, bronchoconstriction, glycogenesis
(2) Increased heart rate, glycogen breakdown, pupil dilation
(3) Increased insulin secretion, lipogenesis, reduced alertness
(4) Decreased blood glucose, increased protein synthesis

■■■

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